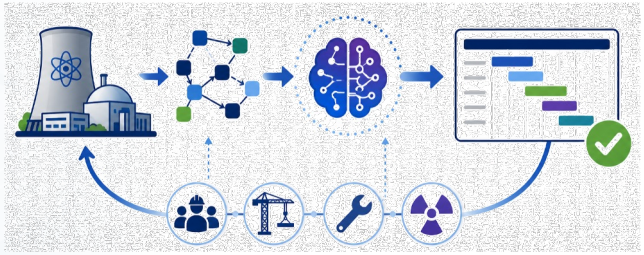
INL/MIS-26-92820



Resource-Aware Scheduling for Nuclear Refueling Outages

Plant Optimization Update and Planning Meeting - June 3rd, 2026

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Battelle Energy Alliance manages INL for the
U.S. Department of Energy's Office of Nuclear Energy

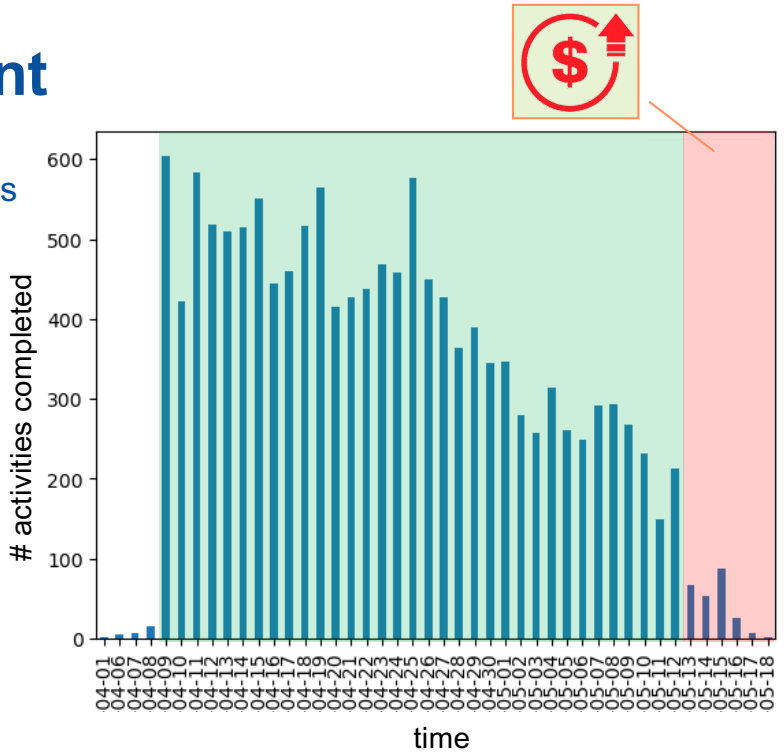
Idaho National Laboratory

Context: Outage Schedule Management

- **NPP refueling outages** are the most complex, high-stakes operations in plant life
- Thousands of interdependent tasks, dozens of specialized crews, shared equipment, strict regulatory gates

The cost of getting it wrong

- Loss of power generation
  - $1–2 million per day
  - A 3-day overrun is a $4.5 million problem
  - Even a 1-day reduction creates an economic value
- Contractor and equipment costs



- **Our principle:** Co-pilot outage managers and staff during outage planning and execution phases
- **Impact**
  1. Increase plant capacity factor (ref. EO 14302 goals, UPRISE)
     - Find opportunities to reduce outage length
     - Reduce risk of outage delays
     - Optimize planned outages to incorporate additional power uprate work without extending outage duration
  2. Reduce workload on outage managers and plant staff throughout outage lifecycle

Collaborators: OPG (?), The Nuclear Company, Constellation

Outage Operational Challenges

Planning phase (> 1 year)

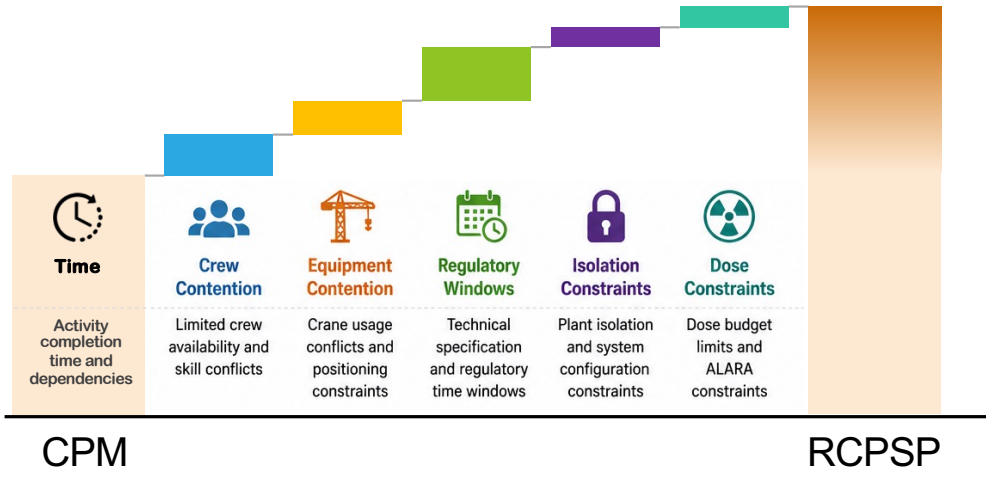
- Build outage scope
  - Define set of activities to be performed
  - Gather data
- Plan
  - Contractors
  - Consumables
  - Equipment
- Pro-actively plan for
  - Unexpected events
  - Bottlenecks and hold points
- Perform “What-if” scenarios

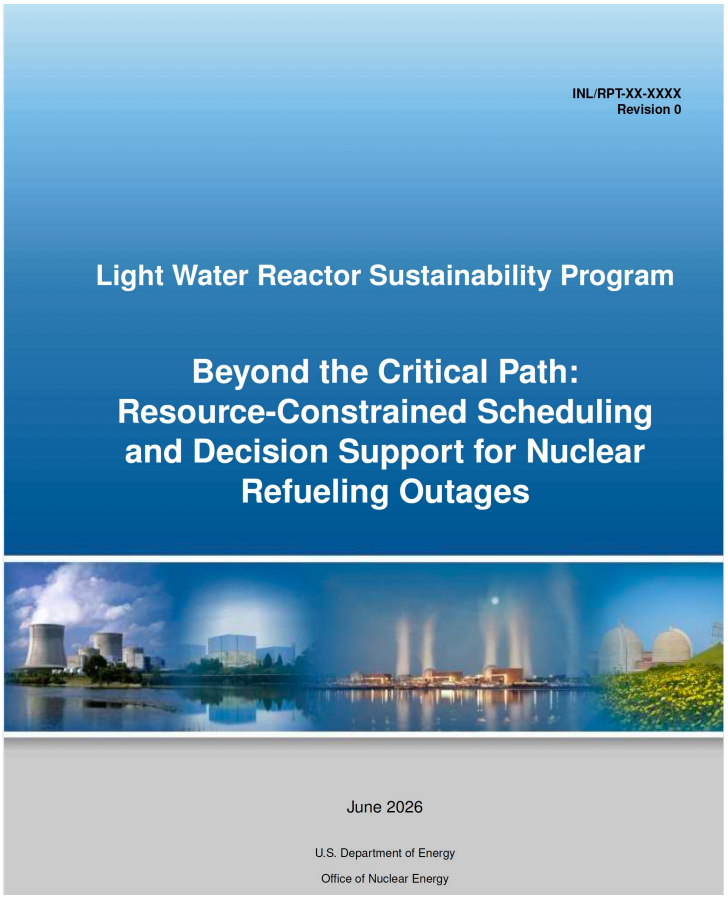
Execution phase (2-4 weeks)

- Manage the unexpected
  - Delays
  - Unplanned activities
  - Change in resource availability
  - Unexpected events
- Handle system state conflicts
- Track outage progress
  - Activity
  - Resources
  - Consumables depletion
  - Dose budgets (ALARA)

FY26 Objectives and Milestones

- Current outage schedules are often dependency-feasible (CPM) but not resource-feasible (RCPSP)
- **Goal:** Develop methods to perform resource constrained optimization
- **Realistic scheduling:** Models every constraint and resource that governs outage execution
- **Tools**
  - LOGOS: https://github.com/idaholab/LOGOS
  - RAVEN: https://github.com/idaholab/RAVEN





FY26 Accomplishments: Outage Simulation Model

- **All constraints enforced, not just task dependency.** The generated schedule is always feasible (i.e., not advisory)
- **The real outage duration, not an optimistic estimate**: The model shows how much longer resource contention makes the outage compared to the CPM lower bound
- **Trade-off analysis built in.** Tasks that can run under different crew configurations (normal, crash, reduced)
  - Planners can compare duration vs. cost before committing
- **22 smart scheduling strategies.** The model evaluates 22 priority rules (including 5 evolved by machine learning) and picks the ordering that best fits the specific constraint mix

Constraint	Examples
Workforce	Skill types, crew counts, shift calendars, overtime limits
Equipment	Asset availability windows, zone-locking to physical locations
Radiation	Cumulative dose budgets per craft; ALARA scheduling
Plant Systems	Mutual-exclusion states (valve alignments, system configurations)
Consumables	Gaskets, specialty seals (finite inventory tracked)
Regulatory	Tec-spec windows, hold points, mode-change gates
Spatial	Zone occupancy limits, concurrent access restrictions
Dependencies	Finish-to-start lags, mobilization lead times

It takes the dependency network build in Primavera P6 and adds the constraint enforcement layer that Primavera often leaves to manual judgment

- It generates a schedule that reflects what crews can actually execute

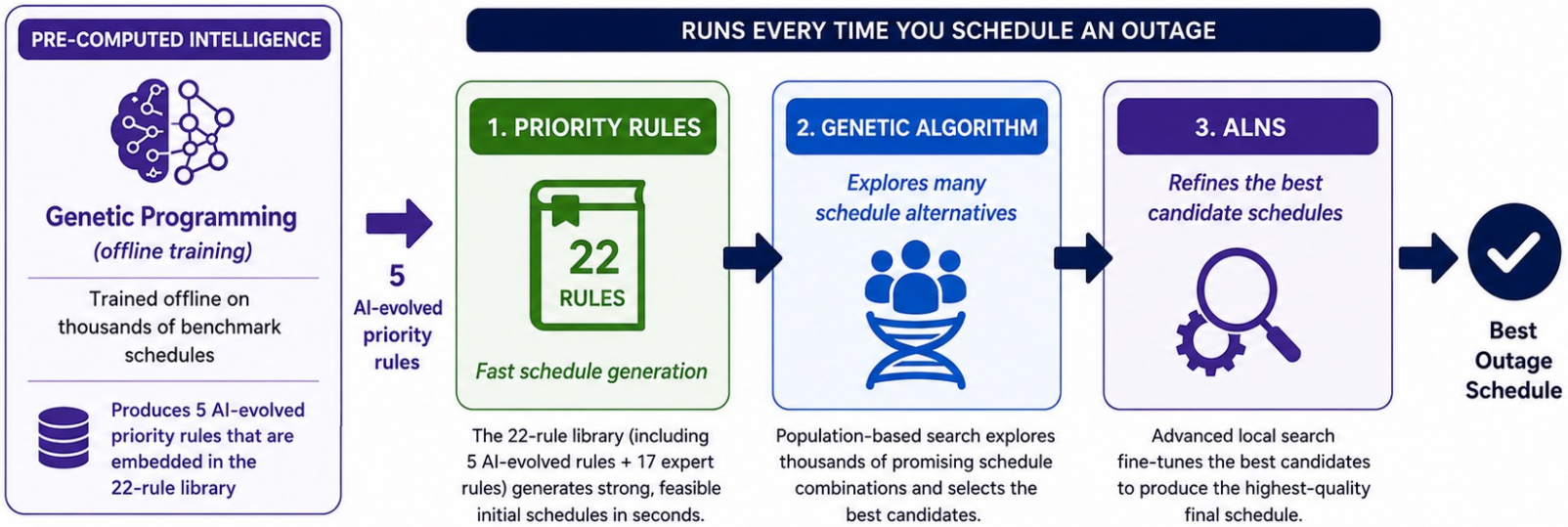
FY26 Accomplishments: Outage Simulation Model

- **Single-call replanning.** When crew counts drop, equipment fails, or an unplanned task appears, one replan call freezes completed work and re-optimizes all remaining tasks in seconds against the current actual constraint state
- **What-if analysis.** Clone the live schedule, test two or three disruption responses side by side, then commit to the best one
- **Know which tasks actually matter.** Color-coded Gantt distinguishes truly critical tasks (any slip extends the outage) from near-critical ones (one disruption away from being critical) from tasks with real flexibility
- **Pinpoint where to add crew.** When the schedule is running long, it identifies exactly which hours and which skill types are the bottleneck (e.g., add two mechanics for 12 hours, not ten mechanics for the whole outage)
- **Schedule margin that can be monitored.** Rather than hidden padding in individual task durations, it concentrates margin into explicit buffers with utilization percentages
- **Time window constraints.** (e.g., shifts, QA or regulatory related)

Capability	Decision Support
Resource-constrained scheduling	What schedule can my crews actually execute?
22 priority rules	Which sequencing strategy minimizes our outage duration?
Fitness score	How close to optimal is this schedule?
Regulatory constraint enforcement	Will we miss any Tech Spec windows?
Multi-mode evaluation	Will a crew surge actually shorten the outage?
CCPM buffers	How much margin do we really have? Where is it being consumed?
Mid-outage replanning	What does our revised schedule look like right now?

FY26 Accomplishments: Schedule Analysis Engine

- **Issue:** With thousands of tasks and constrained resources, the order in which work is started determines whether you finish on time (or not…)
- **Solution:** Heuristic optimization framework for RCPSP problems



Value: The optimization engine evaluates thousands of feasible scheduling alternatives and identifies shorter outage schedules while respecting operational constraints (no manual trial and error)

FY26 Remaining Work

- **Outage simulation model**
  - Code house-cleaning
  - Finalize regression testing
  - Finalize benchmarking
  - Deploy within the LOGOS computational framework

- **Metaheuristic optimization**
  - Scale GA and ALNS to large schedule instances (i.e., 10K-20K activities)
    - Decomposition-based GA for large-instance scalability
    - Hybrid GA + ALNS neighborhood search

- **Publications**
  - PHM conference
  - PSAM conference
  - INFORMS conference

FY27 Plan: Risk-Aware Outage Management Platform

- Decision support platform to: predict probability of outage extension (e.g., from activity completion time uncertainty), quantify schedule risk (e.g., unexpected and unplanned events), and recommend mitigation actions before delays occur

1. Knowledge extraction from plant CMMS and outage databases. Goal is to assess activity duration distributions, resource availability patterns, procurement issues, and historical unexpected and unplanned event frequencies
   - Extract real performance data from plant records
   - Turning decades of fleet experience into quantified uncertainty inputs
2. Extend RCPSP engine to propagate duration (and resource) uncertainties to produce probability-of-extension estimates and evaluate risk scenarios
3. Develop a decision-support interface that propagate data uncertainties and presents risk-ranked schedule alternatives, buffer consumption trends, and disruption recovery options
   - Provide outage coordinators with a risk prognostic dashboard instead of a single deterministic Gantt chart

Vision

FY24–25 Learn	NLP-based extraction of activity duration distributions from historical outage data and outage robustness and resilience metrics
FY26 Schedule	Resource-constrained scheduling engine and schedule optimization methods
FY27 Integrate	Risk-aware outage management platform: From constraint-enforcing scheduler to uncertainty- and risk-informed decision support
FY28 Optimize	Stochastic optimizer: determine schedules that minimize risk and probability of extension (not just nominal duration)
FY29 Automate	Agentic (perceive, reason, suggest) decision support integrated with plant work management systems